Part D — Self-supervised & representation learning

ml-ladder writeup · part D

2026-09-17

Contents

D1 — SimCLR **2**
The math 2

D2 — Masked autoencoder **6**
The math 6

D3 — Probing, and the question that matters **8**
The math 8

What part D settled **11**

Status: complete (D1–D3) · **Constraint:** CIFAR-10 subsets, short budgets; every arm shares data, schedule and seed
Numbers: seeds 0–2, reference seed 0 · raw values in `data/d-metrics.json`
Reproduce: `ladder run D1 ... D3` (needs `uv sync --extra torch`), or `uv run python tools/writeup_figures.py D`
Hardware: swept on a Colab T4

Every part so far learned from labels. Part D removes them and asks what is left to learn from — and the answer is that **the supervision was always in the data; the labels were just a convenient way to point at it.** Two methods point at it differently: SimCLR declares that two augmented views of one image mean the same thing, and MAE declares that a hidden patch should be predictable from its neighbours. Both then get judged the same way, by whether a linear classifier can read the result.

The rungs also stop being independent here. D1's encoder is B4's residual network with the classifier taken off, D2's transformer block is C3's applied to image patches, D1's feature plot is A3's PCA, and D3's probe is D1's. That is the point of a ladder rather than a list.

Notation: z are L2-normalized projections on the unit sphere, τ the contrastive temperature, m the mask ratio, and ϕ a frozen encoder whose features a probe reads.

D1 — SimCLR

Question. With no labels at all, how much of a useful representation falls out of "these two crops are the same picture" — and what does that claim depend on?

The math

NT-Xent is a $(2N-1)$-way classification problem. For a batch of N images, each producing two views, the loss for the pair (i, j) is

$$\ell_{i,j} = -\log \frac{\exp\big(\mathrm{sim}(z_i, z_j)/\tau\big)}{\sum_{k \neq i} \exp(\mathrm{sim}(z_i, z_k)/\tau)}, \qquad \mathrm{sim}(u, v) = u^\top v.$$

Read the denominator: it runs over *every other view in the batch*, so the network is being asked "which of these $2N-1$ is your partner?" The $k \neq i$ is load-bearing — a view's similarity to itself is 1 by construction, and leaving it in the denominator lets the model win by doing nothing.

Why the batch size is a hyperparameter of the objective, not just of the optimizer. NT-Xent is InfoNCE, which bounds mutual information:

$$I(X_i; X_j) \;\geq\; \log K - \mathcal{L}_{\text{InfoNCE}},$$

with K the number of candidates. The bound cannot exceed $\log K$, so a small batch caps how much the objective can ever say — the reason the SimCLR paper needed batches in the thousands and why this rung's numbers are modest.

What the temperature does. The gradient of ℓ_i with respect to a negative z_k is proportional to its softmax weight $p_k \propto \exp(z_i^\top z_k/\tau)$. Small τ sharpens that distribution onto the *hardest* negatives (the ones already closest); large τ spreads it uniformly. So τ is not a scale factor — it chooses

how much of the learning signal comes from near-misses. Asymptotically the loss splits into an **alignment** term (positives close) and a **uniformity** term (features spread over the sphere), and τ sets their balance.

Why the projection head is thrown away. The loss forces *its output* to be invariant to the augmentations — colour, crop, grayscale. Anything the augmentation destroys is information the head must discard to score well. The layer below it is under no such pressure, so it keeps colour and pose information that a downstream task may want. Probing the encoder rather than the projection is not a detail; it is the difference between reading the representation and reading the objective.

What augmentation strength really controls. The positive pair *defines the invariance being learned*. At strength 0 the two views are identical, the task is solvable by any injective map, and nothing is learned about images at all — which makes it the control arm this rung needs.

What I built. A ResNet encoder (B4's `build_network` with the classifier removed), a two-layer projection head, and an augmentation pipeline written as tensor ops rather than torchvision transforms so that the whole batch is augmented on the GPU in one go — random resized crop, horizontal flip, colour jitter, grayscale — with a single `strength` knob scaling all of them and `strength=0` collapsing to the identity. NT-Xent is nine lines; the one that matters is `similarity.fill_diagonal_(float("-inf"))`, and the positive targets are `(arange(2N) + N) % 2N`, which pairs view i with view $i + N$ in both directions. Every arm is probed the same way: encoder frozen, a linear head trained on 4000 labelled images, scored on 2000 test images.

Results (CIFAR-10, 8000 unlabelled images, 600 steps, batch 128; medians over seeds 0–2, ranges in brackets):

arm	probe accuracy
untrained encoder (control)	0.256 [0.245, 0.263]
$\tau = 0.1$	0.359 [0.332, 0.360]
$\tau = 0.5$	0.339 [0.328, 0.347]
$\tau = 1.0$	0.318 [0.314, 0.319]
strength 0.0 (no augmentation)	0.316 [0.303, 0.319]
strength 0.5	**0.430** [0.406, 0.438]
strength 1.0	0.339 [0.328, 0.347]

(The $\tau = 0.5$ and strength-1.0 rows are the same arm — the temperature sweep runs at full strength and the strength sweep at $\tau = 0.5$ — so they share a number by construction, not by coincidence.)

Temperature behaves as the algebra says: monotone in $1/\tau$ over the range swept, hard negatives helping. Augmentation does not.

What surprised me. Three things, and the last is a lesson about metrics rather than about SimCLR.

The augmentation curve is not monotonic. The story everyone tells — stronger augmentation means a harder pretext task means better features — predicts $0.0 < 0.5 < 1.0$. What happened is 0.316, **0.430**, 0.339: the middle wins by nine points over full strength, on every seed — `best/strength` is 0.5 on all three, so unlike D2's mask ratio this optimum is stable, not a coin flip. The explanation is in the definition of the objective rather than in any tuning detail. The positive pair *is* the invariance being learned, so augmenting harder does not simply make the task harder — it changes what

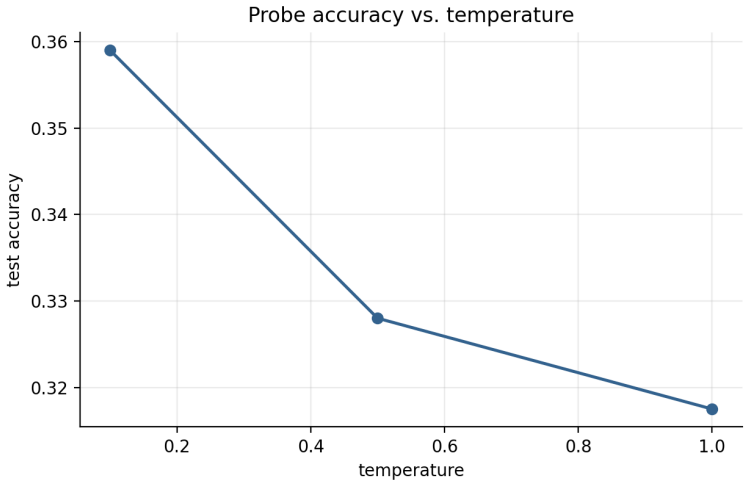


Figure 1: Temperature ablation

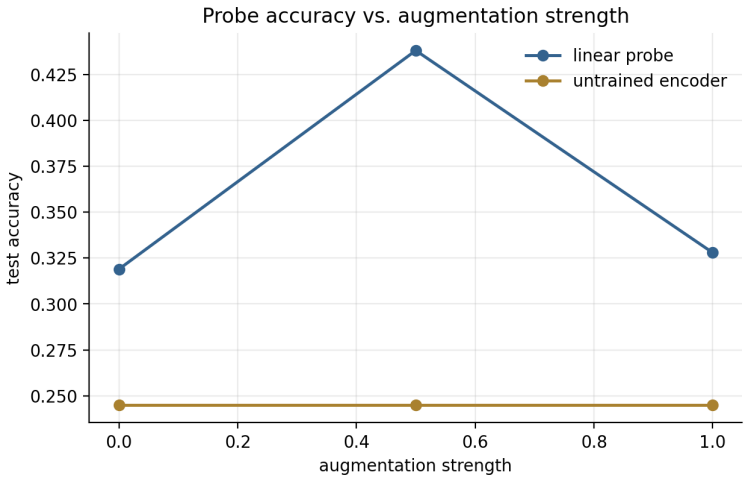


Figure 2: Augmentation ablation

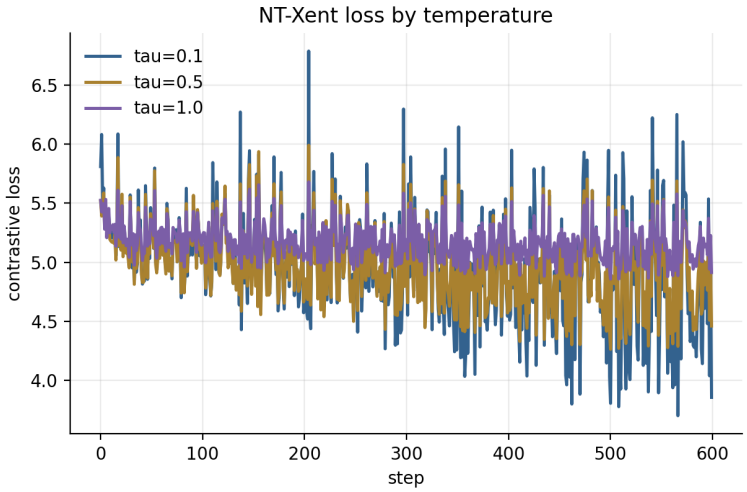


Figure 3: Contrastive loss

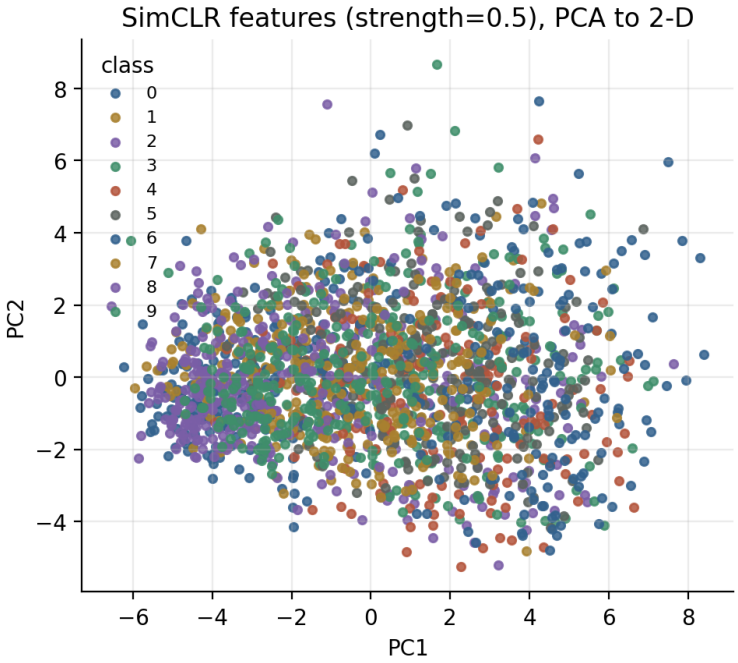


Figure 4: Feature space

the network is told to ignore. At full strength the crops are aggressive enough that two views of one image frequently share no object at all, and the only way to call them the same is to discard the object. The loss still goes down; it is being minimized by throwing information away. This is the "false positive" failure that BYOL and MoCo's momentum encoder were partly designed around, and it is visible at this budget with three arms.

The control arm carries most of the gain, except where it doesn't. Strength 0.0 means both views are identical, so the positive pair is trivially identifiable and nothing about images need be learned — yet it probes at 0.316 against the untrained encoder's 0.256. Compare it to the full-strength arm at the same temperature (0.339) and the arithmetic is uncomfortable: of that arm's +0.083 over an untrained network, **+0.060 is already there with no augmentation at all**. Seventy-two percent of what looks like contrastive learning is what happens to a random network when you push 8000 images through it under any objective that keeps its features from collapsing. Only the strength-0.5 arm escapes this — +0.168 over untrained, of which +0.119 is genuinely augmentation. Without the control arm, both would have been reported as SimCLR's contribution without distinction.

And a metric I had to fix. The rung's first version tracked its best encoder only through the temperature loop, so it reported `best/probe_accuracy` = 0.359 while the strength sweep held 0.430 — a summary that quietly excluded half of the experiment, and a PCA figure drawn from the wrong arm. `augmentation_gain` was worse: defined as strongest-minus-weakest, it reported +0.031 for an effect that is really +0.119 (best arm over the no-augmentation control), *because* the curve is not monotonic. A difference between two endpoints is only a measure of an effect if the effect is monotone, and the interesting result here is exactly that it is not. Both are fixed; the numbers above are from the corrected run.

Open threads. Batch size, which the InfoNCE bound says is a property of the objective — $\log K$ at $N = 128$ is 5.5 nats, and the SimCLR paper used 4096. A finer strength grid to locate the optimum rather than bracket it. And the obvious follow-up to the false-positive story: MoCo's queue, or BYOL, which drops negatives altogether and so cannot be broken this way.

Reading: SimCLR; MoCo; Wang & Isola, *Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere*.

D2 — Masked autoencoder

Question. SimCLR needs a hand-designed augmentation pipeline to say what "the same image" means. MAE needs only a mask. How much should it hide?

The math

The objective is a denoising autoencoder with masking as the corruption. With M the set of hidden patch indices,

$$\mathcal{L} = \frac{1}{|M|} \sum_{i \in M} \|\hat{x}_i - x_i\|^2 ,$$

computed on hidden patches *only*. Scoring the visible ones too would reward copying the input, which a wide enough decoder does perfectly while learning nothing.

Why a high mask ratio, in one argument. Natural images are spatially redundant: a patch is well predicted by its neighbours. At low m the reconstruction is solvable by local interpolation — a smoothing filter would do — and the encoder learns texture statistics. As m grows, the nearest visible patch moves further away, interpolation stops working, and the only remaining route to a low loss is to infer *what the object is* and render the missing part from that. The mask ratio is therefore a dial between a low-level task and a semantic one, which is why the MAE paper's answer (75%) is so much higher than BERT's 15%: language is far less redundant than pixels.

The asymmetry is what makes it cheap. The encoder runs on visible patches only, so its sequence length is $(1-m)N$. Self-attention costs $O(L^2)$, so masking 75% cuts the encoder's attention cost by roughly

$$\left(\frac{(1-m)N}{N}\right)^2 = (1-0.75)^2 = \frac{1}{16},$$

and the decoder — which does see all N positions — is deliberately small and thrown away afterwards. Hiding more of the image makes pretraining *faster*, not slower, which is the opposite of the intuition and the reason the design won.

Positions matter twice. Mask tokens are identical by construction, so without a positional encoding added before the decoder there is nothing to distinguish "the missing patch in the corner" from "the missing patch in the middle" — the model would be predicting an unordered bag of patches. C3's permutation-equivariance argument, arriving in a second domain.

What I built. `patchify/unpatchify` as pure reshapes, and a ViT encoder that is literally C3's transformer block — obtained with `type(CharTransformer(...).blocks[0])` rather than a copy-paste, so the two rungs cannot drift apart. The encoder sees only the visible patches: masking is a `torch.gather` on the patch axis, the mask token is `scatter`ed back before the decoder, and positional embeddings are added on both sides. The loss is

```
(error.mean(dim=-1) * mask).sum() / mask.sum().clamp(min=1)
```

— a masked mean, with the `clamp` there because a degenerate mask of zero patches would otherwise divide by zero rather than fail loudly. `represent()` runs the encoder on *all* patches (no masking) and mean-pools, which is what the probe and D3 consume.

Results (CIFAR-10, 8000 images, 4×4 patches → 64 tokens, 600 steps; medians over seeds 0–2):

mask ratio	final masked MSE	probe accuracy
0.25	0.268	0.349 [0.320, 0.350]
0.50	0.343	0.351 [0.328, 0.356]
0.75	0.480	**0.356** [0.345, 0.362]
0.90	0.676	0.345 [0.325, 0.350]

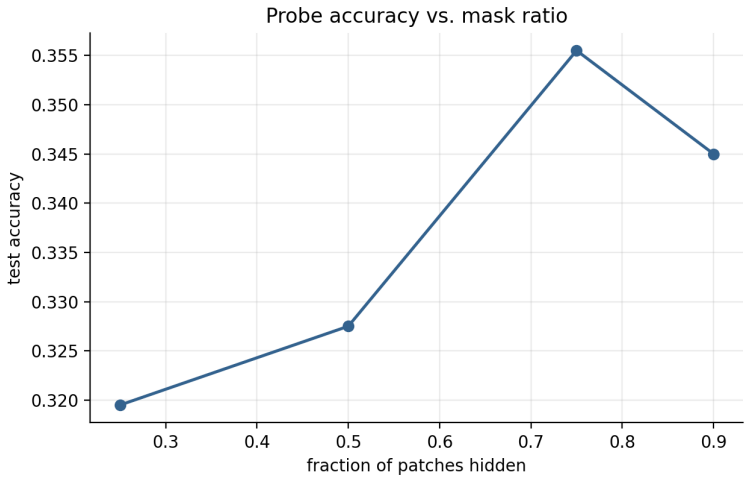


Figure 5: Mask ratio ablation

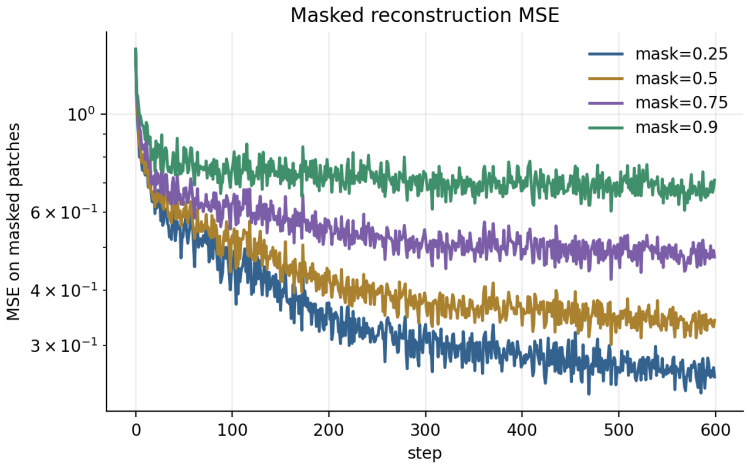


Figure 6: Reconstruction loss

What surprised me. How little the mask ratio mattered, and how easy it would have been to claim otherwise. The median ranking reproduces the MAE paper's answer exactly — 75% on top — but the whole sweep spans **0.7 percentage points**, and the per-seed spread of a *single* arm is twice that. Run `best/mask_ratio` across seeds and it comes out 0.75, 0.75, 0.5: the winner is not stable, and a single-seed run of this rung would have let me write "75% is best, as published" with a straight face. Three seeds turn that into the honest claim, which is that at this budget the mask ratio does not decide anything between 0.25 and 0.9.



Figure 7: Reconstructions

The reconstruction loss, meanwhile, rises monotonically and steeply with the mask ratio — 0.268 to 0.676, a factor of 2.5 — while probe accuracy stays flat. That divergence is the point of the rung: **the pretext loss is not the quantity you care about.** A model reconstructing 25% masking beautifully has learned interpolation, and one reconstructing 90% masking badly has been forced to guess at content. Comparing two self-supervised runs by their training loss compares how hard their tasks were, not how good their features are.

The reconstruction figure makes the same point visually: at 75% the outputs are blurry, low-frequency, plainly wrong in detail — and that model's features probe best.

Open threads. The published result needs scale to appear — ViT-Large, 800 epochs, full ImageNet — so the real thread is whether the flatness here is a budget artefact or a small-data one, which a longer schedule at fixed data would separate. Normalized-pixel targets (the paper's variant) are a two-line change. And the asymmetry argument predicts wall-clock should *fall* as the mask ratio rises; at 64 tokens attention is not the bottleneck, so this rung cannot see it.

Reading: Masked Autoencoders Are Scalable Vision Learners; BERT (the 15% that MAE argues against); ViT for the patch embedding.

D3 — Probing, and the question that matters

Question. Self-supervised learning is not sold on beating supervised training with plenty of labels. It is sold on the regime where labels are scarce. Where is the crossover?

The math

A linear probe measures linear separability, and nothing else. With ϕ frozen, fitting $\min_W \sum_i \ell(W\phi(x_i), y_i)$ is convex in W, so the number it returns is a property of the representation rather than of the optimizer that found it — which is exactly why it is the standard measure.

k-NN measures geometry instead. A cosine k-nearest-neighbour vote trains nothing:

$$\hat{y}(x) = \text{mode}\left\{ y_i : i \in \text{top-}\, k_j \frac{\phi(x)^\top \phi(x_j)}{\|\phi(x)\|\|\phi(x_j)\|} \right\}.$$

The two can disagree, and the disagreement is informative: a representation with a high probe and a low k-NN score is linearly separable but poorly organized — classes distinguishable by

some projection, yet not clustered. Anything built on nearest-neighbour retrieval will fail on such features even though the probe looks healthy.

Label efficiency is the actual claim. Plot accuracy against the number of labels n per class. Supervised training must go through n; a frozen representation only fits a linear head on n. Pretraining's promise is a *leftward shift* of this curve, and its honest failure mode is that the curves cross: with enough labels, learning the features for the task beats reusing features learned for a different objective.

What I built. Four encoders — untrained, SimCLR (D1's `pretrain`), MAE (D2's), and the same architecture trained with labels — all built from the same seed, all fed the same 8000 unlabelled images, all read on the same 2000 test images. Labelled subsets are stratified (exactly n per class) and drawn from one seeded RNG per budget, so every method sees *identical* labels rather than merely equal numbers of them. Features for the k-NN read are computed once per encoder and cached, because otherwise this rung is sixteen forward passes over the pool.

Results. Linear probe accuracy, medians over seeds 0–2:

labels/class	random	SimCLR	MAE	supervised*
5	0.158	**0.250**	0.232	0.577
25	0.186	**0.273**	0.265	0.607
100	0.235	**0.322**	0.317	0.606
400	0.271	0.350	**0.368**	0.608

k-NN accuracy on the same frozen features, same subsets:

labels/class	random	SimCLR	MAE	supervised*
5	0.165	**0.196**	0.169	0.423
25	0.223	**0.270**	0.226	0.600
100	0.257	**0.305**	0.267	0.611
400	**0.305**	0.340	0.293	0.609

*** the supervised column is not a fair comparison, and reading it as one is the mistake this rung taught me** — see below.

What surprised me. Three things, in increasing order of how much they cost me.

The probe and the k-NN disagree, exactly where the objectives say they should. SimCLR's two readings track each other (0.350 probe, 0.340 k-NN at 400 labels). MAE's diverge: it **wins the probe** at 400 labels — 0.368, the best self-supervised number in the part — and simultaneously scores 0.293 on k-NN, *below the untrained encoder's 0.305*. So MAE's features are linearly separable without being metrically organized: a classifier can find the classes, nearest neighbours cannot. This is not mysterious, it is the loss functions. NT-Xent optimizes cosine similarity directly, so cosine k-NN is reading the very quantity that was trained. MAE's loss is a pixel MSE that never once mentions distances between images, so nothing shapes the metric and the probe is free to use whatever directions it likes. Anything you would build on retrieval — nearest-neighbour search, clustering, a memory bank — would work on the SimCLR features and fail on the MAE features, while a probe-accuracy leaderboard would tell you MAE was better.

Both methods beat their own architecture, and by less than the architecture beats nothing. At 5 labels per class, SimCLR is +0.092 over the untrained encoder and MAE +0.070 — real and consistent

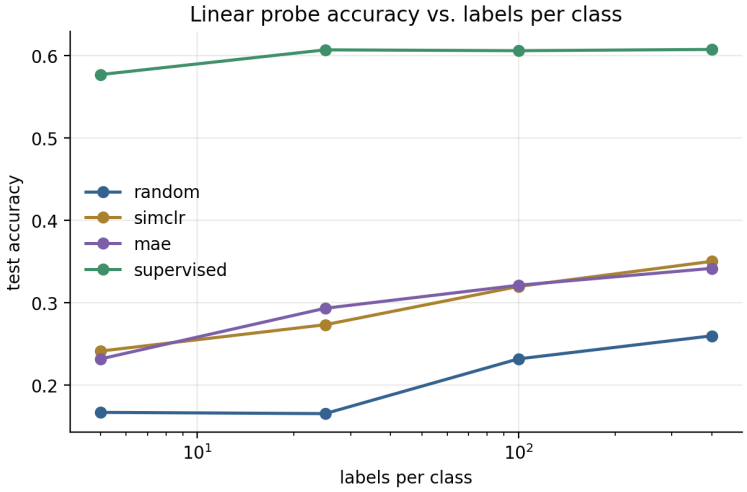


Figure 8: Label efficiency

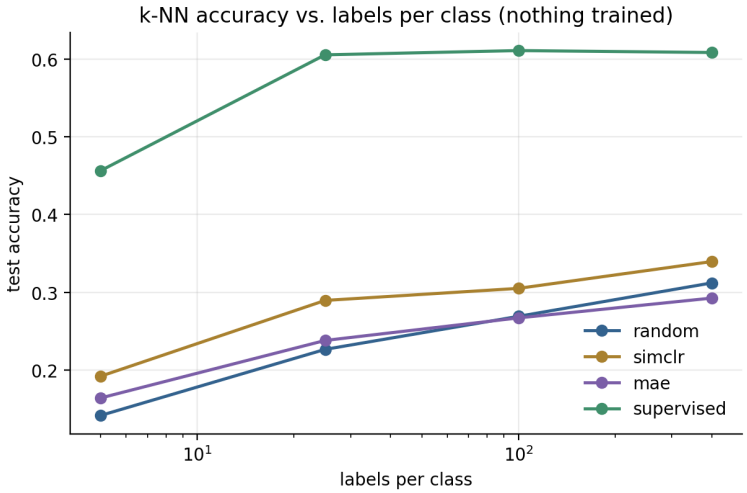


Figure 9: k-NN efficiency

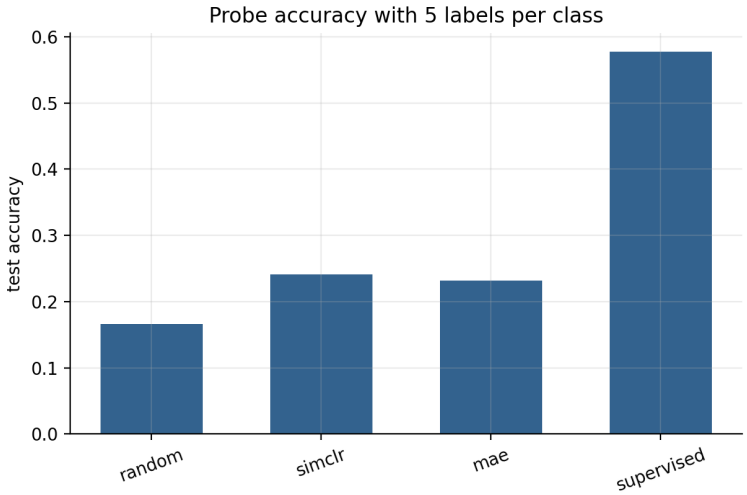


Figure 10: Low-label regime

across seeds, and the gap narrows as labels grow (the D1 and D2 curves converge toward the random one's slope). That is the expected shape: pretraining is worth most where labels are scarcest.

The supervised baseline I wrote does not measure what I built it to measure. It trains the encoder on the **full 5000-label pool** and then probes it with n labels. So its 0.577 at 5 labels per class is not "supervised learning with 50 labels" — it is an oracle that has already seen a hundred times the labels the row claims, and the probe is only re-reading features that were fitted with those labels. That is why the column is flat: 0.577 at $n = 5$ against 0.608 at $n = 400$, a 3-point range where every other method moves by 10 to 12. Every column in the table answers "how good are these frozen features at n labels?" except the last, which answers a different question. The comparison SSL is actually sold on — pretrain on unlabelled data, then fine-tune on n labels, against training from scratch on those same n labels — is **not in this table at all**, and I only noticed because the flatness of the column looked wrong. The derived `simclr_minus_supervised@min_labels = -0.328` is therefore a number about my experimental design, not about SimCLR.

Open threads. The obvious one is the baseline the last paragraph is missing: train the same encoder from scratch on exactly the n labelled images, per budget, and re-plot. Then fine-tuning instead of freezing, which is how these representations get used in practice and which C4 already showed is worth another several points. And a pool much larger than the label set — 8000 unlabelled images against 5000 labelled ones is not the regime where self-supervision is supposed to win, and the ratio is the whole argument.

Reading: SimCLR §B for the probe protocol; MAE §4 on probe vs. fine-tune; MoCo for the k-NN monitor this rung's second table imitates.

What part D settled

1. **The pretext loss is not the metric.** D2's reconstruction MSE rises by a factor of 2.5 across the mask sweep while probe accuracy moves by 0.7 points, and the arm with the *worst* reconstructions has the best features. Any comparison of self-supervised methods by their training losses is comparing how hard the tasks were.

2. **How you read a representation decides what you conclude about it.** MAE wins the linear probe at 400 labels and loses to an untrained network on k-NN with the same features. The two numbers are not redundant: one measures linear separability, the other metric structure, and the objective decides which you get. SimCLR optimizes a cosine similarity and scores well on both; MAE optimizes pixels and shapes no metric at all.

3. **The invariance is the hyperparameter.** Augmentation strength is usually discussed as a regularization knob, but in a contrastive objective the positive pair *defines* what the network is told to ignore. That is why the strength curve is non-monotonic — 0.316, 0.430, 0.339 — and why the failure at full strength is information destruction rather than underfitting.

4. **Two of the three summary metrics I wrote for this part were wrong, and both were wrong in the same way: they assumed a shape the data did not have.** `augmentation_gain` compared the endpoints of a curve that turned out to have its optimum in the middle, reporting +0.031 for a +0.119 effect. The supervised column of D3 assumed a comparison that its own code does not perform. Part B's `usable_lrs` failed the same way. The pattern is now clear enough to name: **a derived metric encodes a hypothesis about the result, and when the hypothesis is wrong the metric does not fail loudly — it returns a plausible number.** The

arm-level measurements were right in all three cases; only the summaries built on top of them lied.

5. **Controls are cheap and they keep moving the answer.** The untrained encoder probes at 0.256 and the no-augmentation SimCLR arm at 0.316 — so at full augmentation strength, 72% of the apparent "SimCLR gain" is delivered by an arm that learns nothing about images. Every part of this ladder so far has had a control arm change the headline number; none of them cost more than a few lines.

6. **This part reproduced exactly, which part C did not.** The sweep was run twice, on two separately allocated T4 sessions, the second time only to pick up D1's corrected summary metrics. Every arm-level number came back identical — D2's ten metrics and D3's thirty-seven bit-for-bit, and D1's seven arms too; the only keys that moved were the three derived ones the fix redefined. So C2's instability was not a general fact about GPU non-determinism, as it was tempting to conclude: it was specific to an all-or-nothing metric sitting on its threshold. Nothing here is near a threshold, and nothing here moved.

Cost of the part: ~1,600 lines including tests, 15 test cases, and a three-seed sweep in about 28 minutes on a T4 (D1 354s, D2 67s, D3 107s per seed). D1 dominates because it trains six encoders from scratch.

Figures regenerate with `uv run python tools/writeup_figures.py D`*; the numbers they are quoted against live in* `data/d-metrics.json`*. The part README is the short version of this document.*